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Simulation in the design and operation of manufacturing systems: state of the art and new trends

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As the industrial requirements change at a rapid pace due to the drastic evolution of technology, the necessity of quickly investigating potential system alternatives towards a more efficient manufacturing system design arises more intensely than ever. Manufacturing systems simulation has proven to be a powerful tool for designing and evaluating a manufacturing system due to its low cost, quick analysis, low risk and meaningful insight that it may provide, improving thus the understanding of the influence of each component. Simulation comprises an indispensable set of IT tools and methods for the successful implementation of digital manufacturing. It allows experimentation and validation of product, process, and system design and configuration. This paper investigates the major historical milestones in the evolution of manufacturing systems simulation technologies and examines recent industrial and research approaches in key fields of manufacturing. It describes how the urge towards digitalisation of manufacturing in the context of the 4th Industrial revolution has shaped simulation in the design and operation of manufacturing systems and reviews the new approaches that have arisen in the literature. Particular focus is given to technologies in the digitalised factories of the future that are gaining ground in industrial applications simulation, offering multiple advantages.

Keywords: manufacturing systems; simulation; information and communication technologies; digitalised manufacturing; review

1. Introduction

In the current highly competitive business environment, product development processes are becoming increasingly complex as products become customised product-service systems (Ong, Yuan, and Nee 2008). The complexity of such systems leads to an inability for analysis by using mathematical methods or to allow realistic models to be evaluated analytically; as such these models are imperative to be studied through the means of simulation (Law 1991; Fowler and Rose 2004). The production system for the realisation of such a system becomes more intricate and complex (Esmaeilian, Behdad, and Wang 2016). Simulation modelling and analysis is conducted in order to gain insight into these complex systems, testing new operating or resource policies and new concepts or systems before implementing them (Chung 2004), and last but not least, gathering information and knowledge without disturbing the actual system (Mourtzis, Doukas, and Bernidaki 2014). The current paper is a continuation of the previous published work of Mourtzis, Doukas and Bernidaki in 2014. The challenges addressed, reflect the evolution of simulation techniques in Manufacturing Systems, over the last five years. In comparison to the previous paper, of 2014, this work investigates the evolution, advances and current technologies of simulation tools. Furthermore, future trends are also depicted, while an extended historical analysis is presented. Admittedly, simulation as part of the design and operation of Manufacturing Systems, has been profoundly reshaped during the last years, providing fertile ground for new research and analysis.

Indeed, with the advent of Industry 4.0, Digitalisation has played a key role to the creation of factories of the future. However, Digitalisation, is a very broad topic, and in order to reflect its actual impact in simulation, further research on cutting-edge, digital, technologies is of great importance. As such, the work presented in this paper, focuses on the analysis of newly emerging technologies, including Cyber-Physical Systems (CPS), Internet of Things (IoT), Cloud Computing, Design of Manufacturing Networks, Augmented and Virtual Reality (AR & VR respectively), Hybrid Simulation and Digital Twins. More specifically, in section 3.2, entitled ‘Product and Production Lifecycle Simulation Tools – Current Status and Future Trends’, several aspects of simulation in product and production lifecycle are examined, whereas the field of research, Design and Planning of Manufacturing Networks, is presented. Correspondingly, in section 3.3, entitled ‘Digitalization in

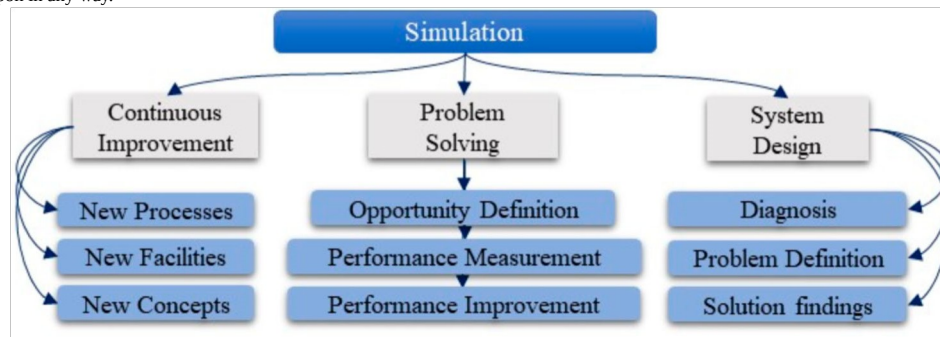


Figure 1. Simulation for design and operation of manufacturing systems.

Simulation – towards the 4th Industrial Revolution’, the digital technologies of Augmented and Virtual Reality, Hybrid Simulation and Digital Twins, as part of the Virtual Factory paradigm are put under the microscope, forming new research topics of Simulation.

Additional to the examination of these modern technologies, a historical analysis is also provided. For the completion of this analysis, a complex literature review algorithm has been applied, in comparison to the keyword-based technique followed in 2014. This literature review algorithm, consists of three phases, which are discussed in detail, in section 1.1, entitled ‘Review Methodology’. Moreover, this approach, has utilised research studies published during the last five years and more keywords. Hence it has yielded more in-depth results, which are illustrated in Figure 2.

In conclusion, the current study, contributes to the body of knowledge, by investigating cutting edge digital technologies and further analysing historical evolution of Simulation, through an extensive review of the available literature, emphasising on studies published the last decade. The findings of this research are discussed in detail in the main body of this paper. Next, a complete section, entitled ‘State of the Market of Simulation Tools in the 4th Industrial Revolution’ has been dedicated to analysing market trends on simulation tools, examining integration of Simulation in Manufacturing. Finally, conclusions are drawn from the presented study and future aspects are discussed, from a higher intuition in Simulation point of view, which in turn will lead to a more accurate approach of the real manufacturing model.

Indicatively, in figure below, the main directions of Simulation added value are presented for Design and Operation of Manufacturing Systems (MS) (Figure 1).

Simulation in the Design and Operation of MS is arguably one of the most multifaceted topics that an Industrial Engineer can face in their career. Simulation is extensively being used as a tool aiming to increase the production capacity. Cost, time, quality, energy efficiency, safety, and productivity are all affected by simulation. This article is focussed towards providing information on the development of Simulation for the Design and Operation of MS accompanied by the new trends as indicated in the literature.

The value of simulation as a tool for the extensive use of modelling and decision support tool is widely recognised especially for highly-complex MS, such as those found in automotive, aerospace, electronics, and semiconductor industries (Fowler and Rose 2004). As Industry moves to the fourth revolution, digitalisation of MS already started from previous eras, is peaking, and creating inter-connected systems that exploit information from different levels of the production. The convergence of industrial production and information and communication technologies is called Industry 4.0. Industry 4.0 includes cyber-physical systems, the Internet of things, cloud computing and cognitive computing (Hermann, Pentek, and Otto 2016).

1.1. Review methodology

Following that, this paper investigates the evolution, the current practices, and the future trends of simulation methods and tools in the context of manufacturing, from the perspective of products, processes, and production. More specifically, CAx, factory layout design, material, and information flow design, manufacturing networks design, manufacturing systems, and networks planning and control, planning and verification (ergonomics, robotics, etc.) are the examined domains. The selection of these domains is made in order to cover the most important functions of a business during the

stages of design, planning, and control on strategic, tactical, and operational levels. Furthermore, emerging approaches and their effect in the current literature are also studied: augmented and virtual reality in product and process design, digital twin and hybrid simulation towards the Factory of the Future Vision.

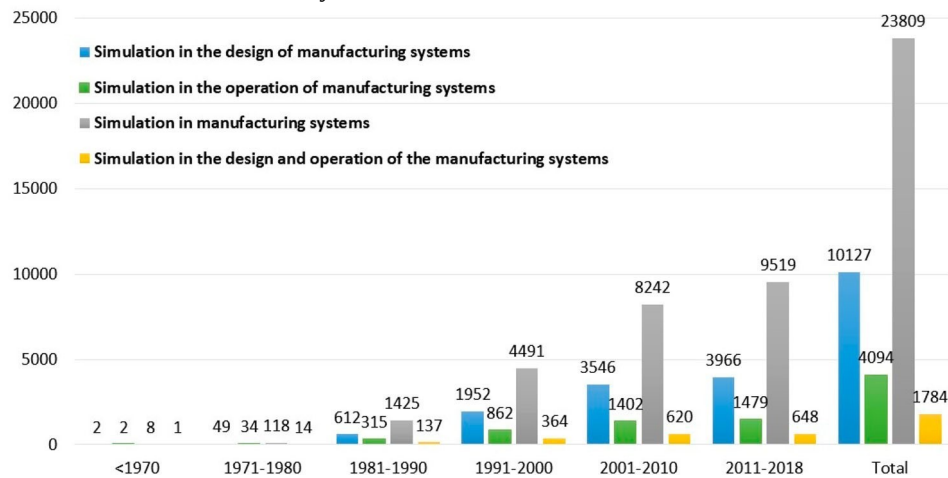


Figure 2. Number of publications based on Scopus database score.

This review is based on academic peer-reviewed publications that use simulation in areas related to manufacturing, over a period of 58 years. The ever-increasing importance of simulation in manufacturing is revealed by the exponential increase of publications on the topic. Before the structured approach, a search in Scopus database by the keywords (a) simulation in manufacturing and (b) simulation in the design and operation of manufacturing systems was performed to quantify the volume of related work in total. It resulted in more than 23,000 publications for the first search and more than 10,000 for the second one. Although there were only a few research publications relevant to simulation in manufacturing systems until 1970 (14), according to Scopus database, a notable growth followed since leading to more than 23,000 publications until 2018. Thus, simulation is an ever-growing part of manufacturing that may support its different aspects. The evolution of the research on this topic through the years is depicted clearly in Figure 2 where the research results have been categorised from 1970 until today. It is clear that an overview and a holistic view are needed in order to extract the key points and give to the reader the latest trends on the domain. More specifically, the outcome of this analysis is that during the last decade there has been a definite increase in simulation related publications. Thus, there is room for further research and development in the field.

Then, the review methodology was carried out in three stages: (a) search in scientific databases (Scopus, Science Direct, and Google Scholar) with relevant keywords for the seven examined domains mentioned before (Searching for Search Field Type Article title, Abstract, Keywords), (b) identification of relevant papers by abstract reading, and (c) full-text reading and grouping into research topics. The three stages procedure is presented in the figure below.

Considering that a lot of review works on simulation have been performed by 2013 (Mourtzis, Doukas, and Bernidaki 2014; Negahban and Smith 2014), our aim is to focalise to the advances on simulation covering the period 2010–2018. The keywords used in phase 1 for the seven examined domains are presented in table 1 below, as well as the relevant results found in total and for the last eight years.

Our search was performed following two formulations, where D_i represents the examined domains:

Total	(TITLE-ABS-KEY (manufacturing AND systems) AND TITLE-ABS-KEY(1 (Simulation) AND TITLE-ABS-KEY (D_i)))	
2010–2018	(TITLE-ABS-KEY (manufacturing AND systems) AND TITLE-ABS-KEY (Simulation) AND TITLE-ABS-KEY (D_i)) AND (LIMIT-TO (PUBYEAR, 2019) OR LIMIT-TO (PUBYEAR, 2018) OR LIMIT-TO (PUBYEAR, 2017) OR LIMIT-TO (PUBYEAR, 2016) OR LIMIT-TO (PUBYEAR, 2015) OR LIMIT-TO (PUBYEAR, 2014) OR LIMIT-TO (PUBYEAR, 2013)) OR LIMIT-TO (PUBYEAR, 2012)) OR LIMIT-TO (PUBYEAR, 2011)) OR LIMIT- TO (PUBYEAR, 2010))	(2)

The presented methodology was carefully designed by the author, leading to the formulation provided above. The formulation is divided into two cases. The first case, is a general search formulation. It is focused on the manuscripts whose title, abstract or keywords contain the term ‘*manufacturing systems*’ while at the same time the term ‘*Simulation*’ and the

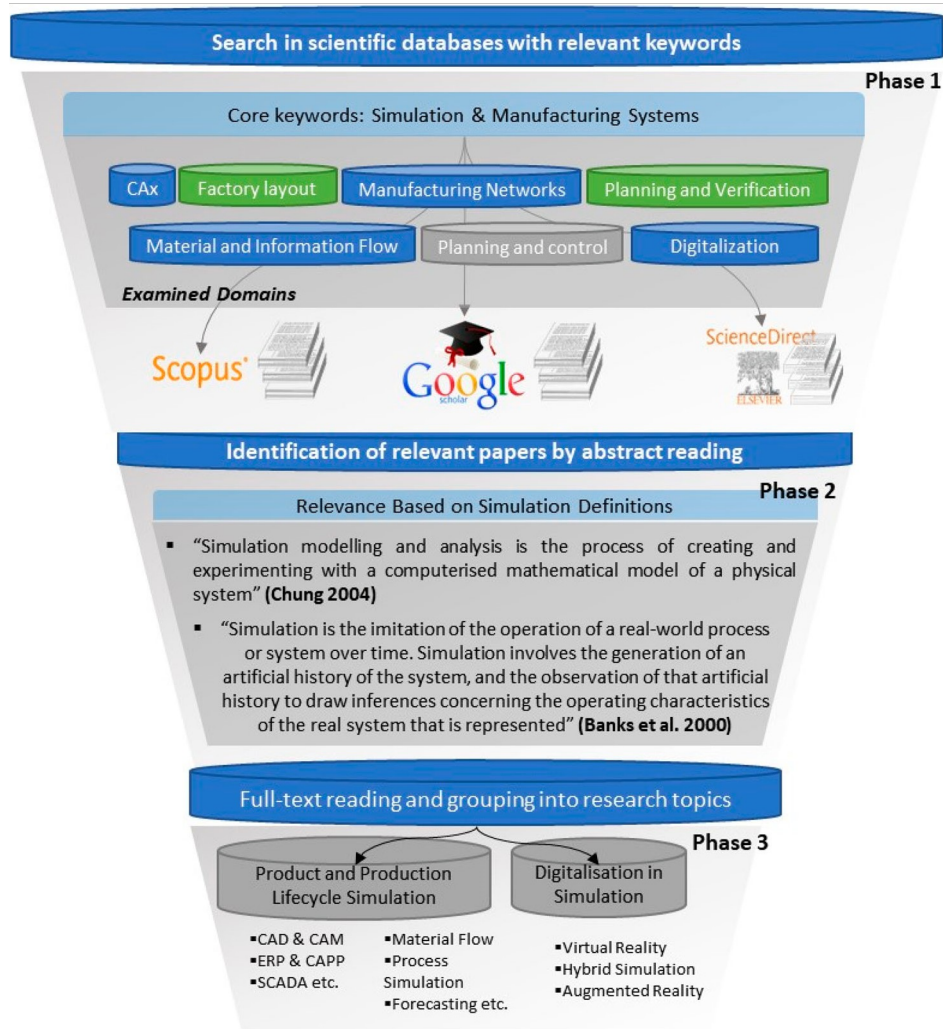


Figure 3. Review methodology.

term ‘ D_i ’ are included also in the areas mentioned above. The second stage of the of the search is focused on manuscripts published from 2010 until 2018.

In Figure 3, the research methodology is presented by the author. The research methodology is a multi-phase process as shown. During the 1st phase of research, the author used popular search engines for academic publications. Continuing towards the 2nd phase of the research the author selected relevant publications by abstract reading. Finally, during the 3rd phase the author, selected relevant work by full-text reading.

The depth of the search is indicated by the extensive time period of 58 years that was examined as well as from the fact that the first search using as keywords ‘simulation in manufacturing’ resulted in 23000 results and the search using the keywords ‘simulation in the design and operation of manufacturing systems’ resulted in 10000 results. The research is divided in two sections, based on the publication year:

- Section 1: contains publications young than 2010
- Section 2: contains publications from 2010 to 2018

The presented work, is a thorough literature review on simulation practices for Manufacturing Systems, focusing on the latest advances.

Scopus is an abstract and citation database provided by ELSEVIER. Thus, it can be considered as trustworthy source of information and statistical data.

It can be easily presumed that the latest years there is an exponential increase in the research on the topic of Simulation in the Design and Operation of MS. The results of Table 1 are also presented in Figure 4 below.

Table 1. List of search terms per domain and results.

	<i>D</i> ₁ _CAX	<i>D</i> ₂ _Factory layout design	<i>D</i> ₃ _Material & information flow design	<i>D</i> ₄ _Manufacturing networks
Keywords	CAD	Digital mock up	Product data management	Supply chain
	CAE	Layout design	Material flow design	Manufacturing strategy
	CAM	ERP	Process simulation	
	CAPP		Knowledge management	
	CIM		Process control	
Total No	2148	76	677	3211
2010–2018	511	24	298	943
	<i>D</i>₅_Manufacturing systems networks planning & control		<i>D</i>₆_Planning & verification	<i>D</i>₇_Virtual factory
Keywords	Manufacturing execution systems		Ergonomics	Augmented reality
	Supervisory control		Robotics	Virtual reality
	Data acquisition			Hybrid simulation
Total No	1864		109	929
2010–2018	440		47	257

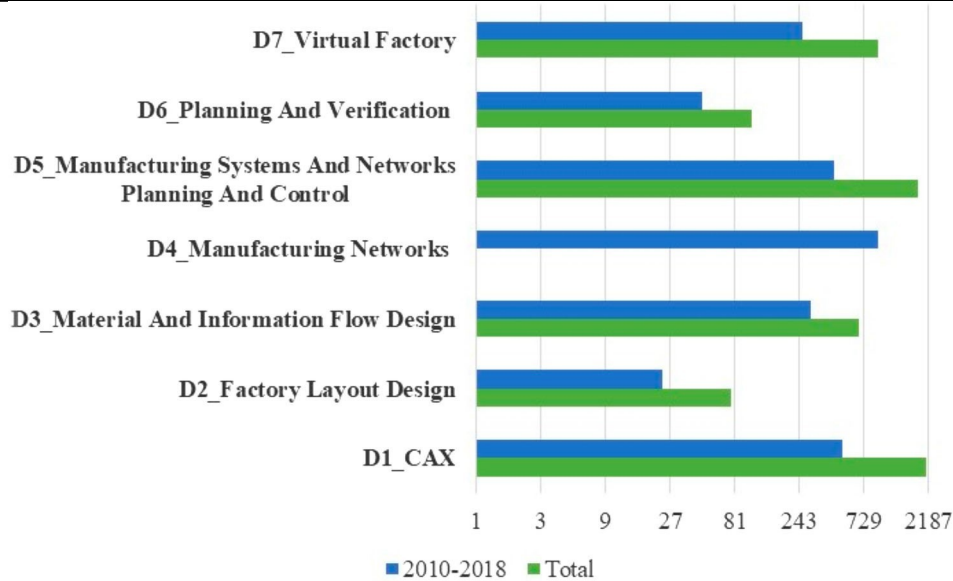


Figure 4. Research finding per domain (*D*) in total and in last 8 years (logarithmic scale).

2. Historical evolution of simulation methods for manufacturing systems design and operation

Following the work of (Goldsman, Nance, and Wilson 2010) and the recent technological advancement, five historical periods could be identified: the pre-computer era from Buffon to World War II starting back on 1777 and closed on 1945, the formative period from 1945 to 1970, the expansion period starting on 1970 and ending on mid-80s, the period of Consolidation and Integrated Environments mid-80s to 2000 and after that the ongoing era of digitalisation starting back on 2000.

Starting back on 1777, it is generally considered that the contemporary meaning of simulation originated by the work of Comte de Buffon who suggested a Monte Carlo-like method in order to determine the outcome of an experiment consisting of repeatedly tossing a needle onto a ruled sheet of paper. It is apparent that although the term ‘Monte Carlo method’ was coined much later in 1947, at the start of the computer era, stochastic sampling methods were used long before the evolution of computers (Seco and Verhaegen 2013). In the mid-1960s, simulation makes a significant leap with the contribution of Tocher and Owen, who developed the General Simulation Program, which is considered to be the first general-purpose simulator. With the use of light, sound, motion, and even smell to immerse the user in a motorcycle ride,

Heilig designed the ‘Sensorama ride’, which is considered to be a predecessor of Virtual Reality (VR) (Lu, Shpitalni, and Gadh 1999). In 1968, Sutherland presents a head-mounted 3D display, setting the foundations for Augmented Reality (AR) (Sutherland 1968). General Precision Equipment Corporation and NASA use analog and digital computers to develop simulators for the Gemini project (Goldsman, Nance, and Wilson 2010). Since the 1970s and 1980s, Nelson and Winter began to define a secure ground and indicate a trajectory for simulation models (Nelson and Winter 1982). Following this tradition, among the numerous significant developments of the evolutionary approach, a ‘first generation’ of simulation models (Windrum 1999) has been increasingly implemented. In 1978, computer imaging with the introduction of digital image generation is a significant contribution to the advancement of simulation. At the beginning of the 1980s, major breakthroughs take place, such as flight simulators for military purposes (Liversidge et al. 1980), naval and submarine simulators are produced, and NASA develops relatively low-cost VR equipment. All that is connected to the Microelectronics evolution on the mid-80s, an enabler for this breakthrough (Suri 1985). However, there are a lot of references in literature, which stand out as key references: Kaizer (Kaizer, Heller, and Oberkamp 2015), the textbooks published by Roache (2009), and Oberkamp and Roy (2010).

In the early 1990s, real-time simulations and interactive graphics become possible due to the increased computing power. Also, commercial VR applications become feasible (Lu, Shpitalni, and Gadh 1999). Kellner, Madachy, and Raffo (1999) substantiated an overview of Software Process Simulation Modelling and identified the simulation objectives, the scope of simulation models and offered guidance in order for appropriate modelling approaches to choose. The earliest frameworks considered are the Capability Maturity Model Integration and the Technology Readiness Levels (National Security and International Affairs Division 1999). Oberkamp, Pilch, and Trucano (2007) along with NASA developed the Predictive Capability Maturity Model, which is a broader application of the original framework of Pilch et al. (2004), on scientific computer simulation (NASA 2008).

Further technological advancement in the last decade concerns software applications enhanced by virtual reality, improved interfaces and utilities, and agent-based modelling, indicating the inclusion of intelligent components in sophisticated software. In this period the first approaches on hybrid simulation were reported in the literature, in seismic loadings simulations, combining numerical and experimental methods (Saouma and Sivaselvan 2014). Moreover, during this period, digital twin approaches also arose in the literature (Tuegel et al. 2011), bringing together real-time data acquisition and sophisticated simulation of alternative scenarios. The fourth industrial revolution takes the automation of manufacturing processes to a new level by integrating multiple digital technologies (Mourtzis, Doukas, and Bernidak 2014), bundling them together under customised and flexible mass production models (WEF 2018).

As the industrial landscape moves toward the fourth industrial revolution, new methods, and approaches that enable simulation in manufacturing emerge, integrating data from various sources and expanding its fields of application (Mourtzis, Papakostas, et al. 2012). Handling complex cases of resilient factories, intelligent maintenance, self-organized adaptive logistics, customer integrated engineering, and smart factory architectures, require the integration of production data in modelling that can only be accomplished with the use of advanced simulation and IT. Moreover, the value of products will be more and more based on their software counterparts and not on their construction or design features under the Product Service Systems (PSS) philosophy (Mourtzis et al. 2018). The historic evolution of simulation is also depicted in Figure 5 below.

3. Simulation in the design and operation of manufacturing systems

The upcoming sections review some of the recent publications. At first, the traditional simulation approaches and their evolution are presented, highlighting state of the art publications that have re-shaped them in the last few years. Then, the advanced methods, which arose from the evolution of technology or from the combination of traditional ones, will assist manufacturing to its transition to the fourth industrial revolution. The abovementioned progress is presented in Figure 6 below.

3.1. The intertwined lifecycles of product and production

Initially, the basic concept or idea for a product is conceived considering the initial request from a customer and, subsequently, it is transformed into a working prototype (Mourtzis et al. 2017). The aim of the product development process is to ensure that the product and its components meet the required specifications (Bodein, Rose, and Caillaud 2013). Thereinafter, the digital factory concept enables the synchronisation of the engineering processes that affect the entire value chain, accessing all product information needed, regardless of their physical location (Kühn 2006). Augmented and virtual reality simulations have arisen as a way to simulate product designs before production (Nee et al.

2012). The product after its launch to the market is supported by complementary services, maximising customers' satisfaction and retrieving valuable information for the manufacturer (Mourtzis, Vlachou, and Zogopoulos 2018).

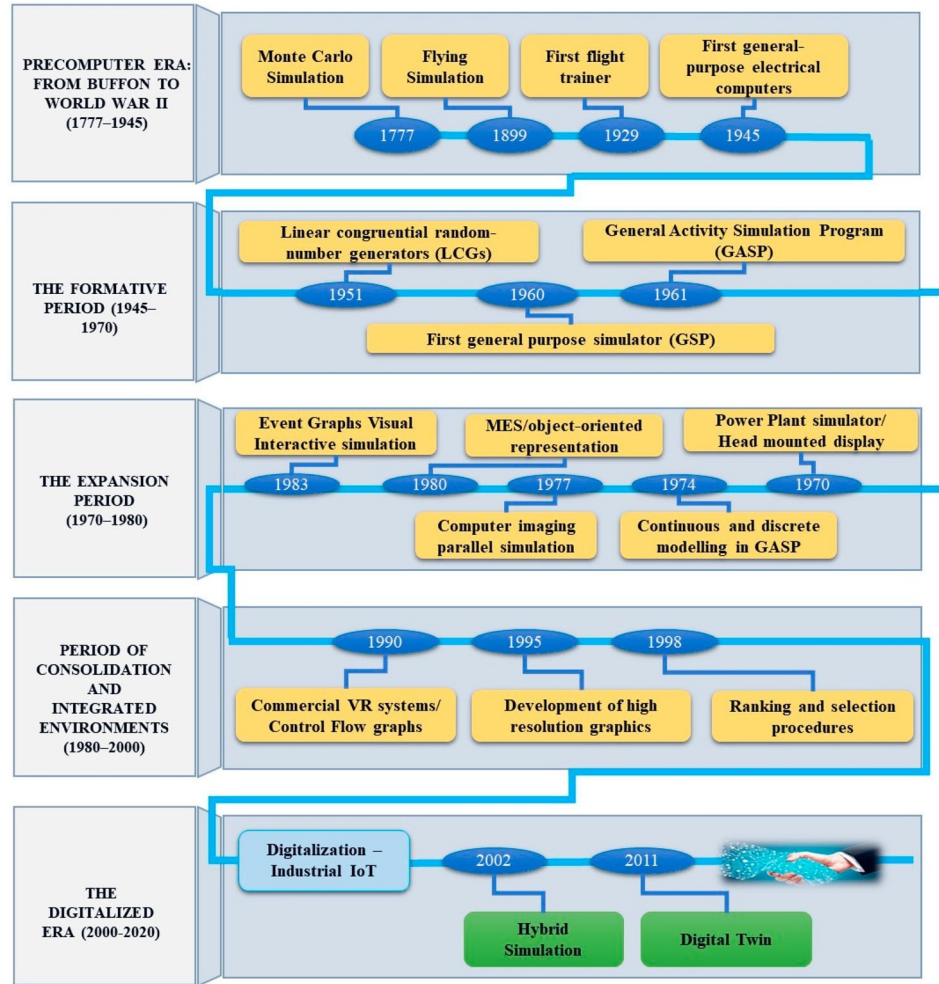


Figure 5. Historical evolution of simulation.

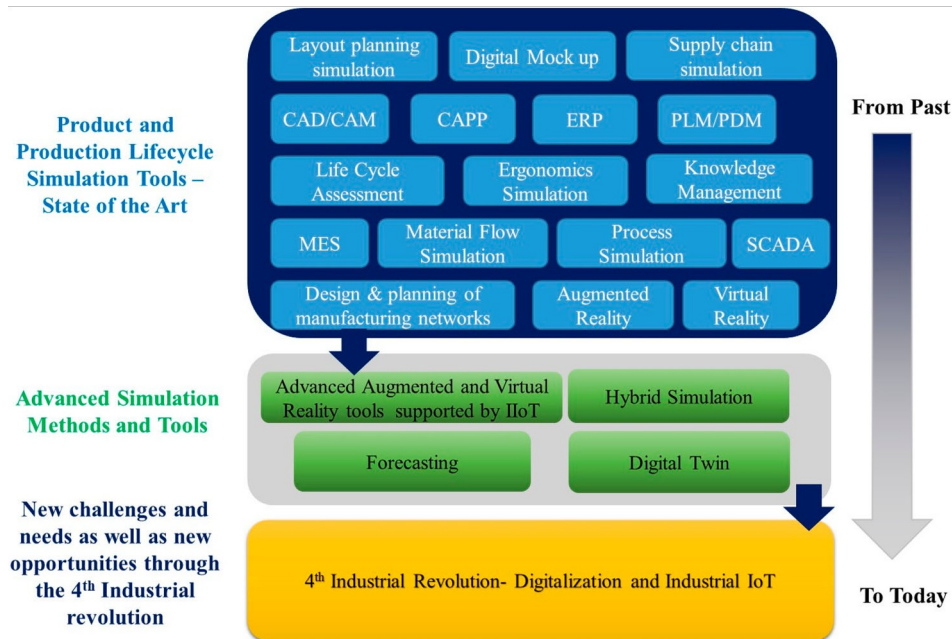


Figure 6. Evolution of simulation.

At the end of the product lifecycle, the product can be recycled or re-manufactured, retaining its geometrical form and reused either for the same purpose or for secondary purposes (Hauschild, Jeswiet, and Alting 2005; Sundin, Tang, and

Mårten 2005). These aspects of circular economy are still under development, as they have low efficiency (McKinsey & Company 2016), and they need to be included in new simulation approaches that will include all the phases of the product lifecycle.

As far as the production lifecycle is concerned, at first, the design stage starts with a stakeholder analysis to identify the constraints and degrees of freedom for the design (Jacobsen et al. 2002). After the specification of the requirements, the design and redesign of manufacturing systems follow, which is summarised as manufacturing system engineering (MSE). During the process development phase, the rough solutions from the design phase are refined to a level that allows investigation with analysis tools. If system dimensions are fixed, the first contact with suppliers is established to integrate them into the further system definition (Mertins, Rabe, and Muller 1998). Two conflicting factors are characteristic of the ramp-up period: low production capacity, and high demand (Terwiesch and Bohn 2001). Finally, in order to address environmental problems, manufacturers make significant efforts to recycle, namely, the recovering of materials or components of used equipment, for the purpose of making them available for new products or processes (Jovane et al. 1993).

The linearity in the lifecycles presented above is an oversimplification. In industrial practice, recursive loops between the phases are necessary, both in the product, and production design and engineering in order to synchronise the different departments involved in the process, and to align mind-sets and procedures.

3.2. Product and production lifecycle simulation tools- current status and future trends

3.2.1. Computer-aided design

Computer-Aided Design (CAD) is the technology related to the use of computer systems to assist in the creation, modification, analysis, and optimisation of a design (Conway, Johnson, and Maxwell 1960). A framework of a collaborative intelligent CAD, which consists of the collaborative design protocol and the design history structure, is proposed in (Lee, Kim, and Banerjee 2010). Simulation in tandem with axiomatic design allowed the creation of systems with the highest probability of satisfying the functional requirements (Melvin and Suh 2002). Bodein, Rose, and Caillaud (2014) presented innovative CAD methods for complex parts modelling in parametric CAD system and their application in an industry context. Moreover, a unified approach was proposed by Salonitis and Stavropoulos (2013) that integrated CAD, CAM, and CAPP systems and address the challenges faced during the process planning phase. Future developments are focused on improving human-computer interface, with more intuitive functionalities, such as gestures and voice commands for instance (Kang et al. 2013).

Leu et al. (2013) reviewed the state-of-the-art methodologies for developing CAD model-based systems for assembly simulation, planning, and training, where methods for CAD model generation from digital data acquisition and a VR/AR system are extensively described. Liu et al. (2015) presented an integrated framework for design, modelling, and optimisation of fiber-reinforced plastic parts. An integrated simulation method for the analysis of the entire injection moulding process in parallel with the quality evaluation of the mould during the early design stage proposed in (Fu and Ma 2019). (Vik et al. 2014) demonstrated a fully integrated software tool which will automatically generate alternatives for the layout of facilities and factories through data exchange between simulation tool and the CAD system.

3.2.2. Computer aided manufacturing

Computer Aided Manufacturing (CAM) can be perceived as the use of computer systems to plan, manage, and control the operations of a manufacturing plant through either direct or indirect computer interface with the production resources of the plant. The application of CAM in production offers advantages to a company to develop capabilities by combining traditional economies in scale with economies with the scope of resulting in the desired flexibility and efficiency (Makris, Mourtzis, and Chryssolouris 2012). Developing algorithms for optimum tool paths determination for free-form surfaces is an ongoing research subject (Lazoglu, Manav, and Murtezaoglu 2009). CAM systems in the digitalised and interconnected era need to focus on collaborative techniques, effective communication, and efficient data exchange (Makris, Mourtzis, and Chryssolouris 2012).

3.2.3. Computer aided process planning

Process planning deals with the selection of necessary manufacturing processes and the determination of their sequences to 'transform' the ideas of designers into a physical component, both economically and competitively (Xu, Wang, and Newman

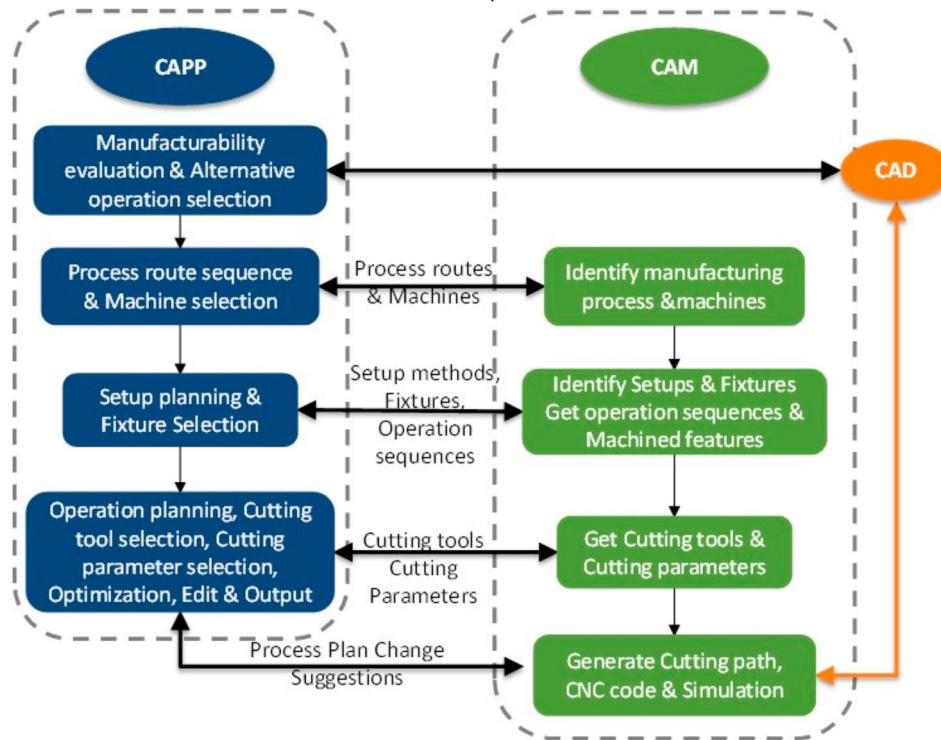


Figure 7. Relations between CAM, CAPP, and CAD systems (adapted from Ming et al. 2008).

2010). Mathematical models of the processes are also required, as well as knowledge about its behaviour, influencing factors, and constraints (Makris, Michalos, and Chryssolouris 2012).

Currently, manufacturing is moving towards a more advanced, flexible, and environmentally-friendly policy and so it demands intelligent CAPP systems (Yusof and Latif 2013). Hamou et al. (2014) developed a CAPP simulation module in order to optimise manufacturing tolerances, overcome stochastic aspects, and complexity, and compute manufacturing dimensions.

CAD/CAPP/CAM systems (Figure 7) are integrated in order to evaluate alternative process plans in different levels, through which, the exploitation of the available resources and their optimal setup can contribute in the overall sustainability of the production facilities (Salonitis and Stavropoulos 2013). Shin et al. (2003) introduce a highly specialised CAPP/CAM integrated system, called Generative Pattern Machining (GPM), for automatic tool paths generation to cut die pattern from the CAD model of the stamping die, currently used by Daimler-Chrysler pattern shop. Wang (2013) developed a web-based service-oriented CAPP system with function blocks for machine availability monitoring and process planning, supporting cloud manufacturing. Though conventional CAPP systems are incapable of adjusting to dynamic operations, thus are found improper or unusable to specific cases or require a significant amount of time and effort (Wang 2013). The significance of the enterprise systems and simulation integration in improving shop floor's short-term production planning capability discussed by Asif Rashid and Benny Tjahjono (2016).

3.2.4. Digital mock up

A digital mock-up (DMU) consists of 3D models, which integrate the mechanical structure of a system in a virtual prototype, reducing time and money and facilitating collaborative product design (Sibois et al. 2013). The developed DMU visualisation systems can import large CAD models into the memory of a computer, while simultaneously being based on a compression representation with triangular patches (Sun 2007). More recently, AIRBUS Military researches the implementation of the industrial Digital Mock-Up (DMU) concept in order to support the industrialisation process of a medium-size aero-structure (Menéndez et al. 2013). Finally, colouring the DMU enables highlighting the required attributes from the customers (Shehab et al. 2010). Future developments will focus on expanding DMUs to design services (Menéndez et al. 2013). Towards the extension of DMU technology to digital manufacturing field (2016) introduced a framework of space optical remote sensor using DMU and based on 'simulation-driven design' the performance and the product quality assurance are checked. The integration of Modelica simulation software with a 3D digital mock-up described by (Matsuda et al. 2017) to translate product concepts to 3D models.

3.2.5. Material flow simulation

Material flow within a manufacturing environment is the movement of materials through a defined process or a value stream within a factory or an industrial unit for the purpose of producing a finished product (Ad Esse Consulting Ltd. 2007).

Fischer et al. (2010) present an automated motion planning framework, integrated into the scene modelling workflow from a material flow simulation framework, which automatically generates motion paths for moving objects. An integrated planning approach for the evaluation and the improvement of the changeability of interlinked production processes before the event actually takes place is proposed by (Albrecht, Faatz, and Abele 2013). An approach to support the management of a ship repair yard by integrating into an open and flexible system a number of critical business functions with production planning, scheduling, and control is presented by Mourtzis (2005). An assignment logic of the workload to the resources of a dairy factory has been implemented on a software system as reported by Mourtzis (2006) simulating the operation of the factory and creating both a schedule for its resources and a set of performance measurements, which enable the user to evaluate the proposed schedule. A P3R-driven (product, process, plant, and resource) modelling and simulation system in Product Lifecycle Management (PLM) is introduced consisting of a data structure for a simulation-model generation, an application-based on an object-oriented model and a concurrent material flow analysis system (Lee et al. 2012).

With regard to material flow simulation, it can be very time-consuming to build and verify large models with standard commercial software. Efficient simulation-model generation will allow the user to simplify and accelerate the process of producing correct and credible simulation models in the future (Lee et al. 2012). The necessity of a more realistic prediction of disturbances in the material handling processes can be the foundation for optimised approaches that combine material flow simulation and physical simulation (Glatt, Kasakow, and Aurich 2018).

3.2.6. Process simulation

A manufacturing process is defined as the use of one or more physical mechanisms to transform the shape of a material's shape and/or form and/or properties (Chrysosolouris 2006). Work in the field of process modelling and simulation is reported as early as in (Toenshoff et al. 1992) and still continues to grow (Brinksmeier et al. 2006). Emerging composite manufacturing processes increase the need for process simulations to reduce the time and cost associated with the product and process developments (Mohan et al. 1998). The FEDES software (Finite Element Data Exchange System) includes case studies for simulation of manufacturing process chains including aero-engine components (Afazov 2013). Paralikas, Salonitis, and Chrysosolouris (2013) developed a methodology for sequentially simulating each step in a manufacturing process. Moreover, in order to empower decision making in process planning, Hurtado et al. (2015) describe an application of modelling and process simulation techniques in order to evaluate the usability of each process.

Process simulation is a highly growing field. Apart from the creation of unified models, the macro-scale manufacturing process chains need to be optimised using numerical techniques such as the FEM. Modelling and simulation of micro-scale manufacturing process chains are still a challenge, as there is little work done in simulation tools. Also, the macro-scale manufacturing processes chains are not fully understood and there are still challenges for improving the manufacturing process chains related to different industries and for developing new manufacturing process chains for new emerging applications (Afazov 2013).

3.2.7. Layout planning simulation

Facility layout planning refers to the (re-) design of the allocation plans of the resources/ equipment in a manufacturing shopfloor that largely affects its performance (Jiang and Nee 2013). Using predefined objects, a layout model can be implemented in 3D, providing the user with the ability to move through factory mock-ups, walkthrough, inspect, and animate motion in a rendered 3D-factory model (Kühn 2006). Moreover, a method implemented on a web-based tool is proposed in (Michalos, Makris, and Mourtzis 2011), focused on human-based assembly systems.

More recently, Michalos et al. (2015) suggest a methodology that enables the user to design and configure a production line against multiple-user criteria. A factory and manufacturing planning, usability, analysis functionalities, and accuracy demands simulation are discussed in (Pentenrieder et al. 2007), introducing augmented reality simulations in the field. Future efforts should focus on supporting increased interoperability between layout and production simulation software, thus providing increased extensibility (Shariatzadeh, Sivard, and Chen 2012).

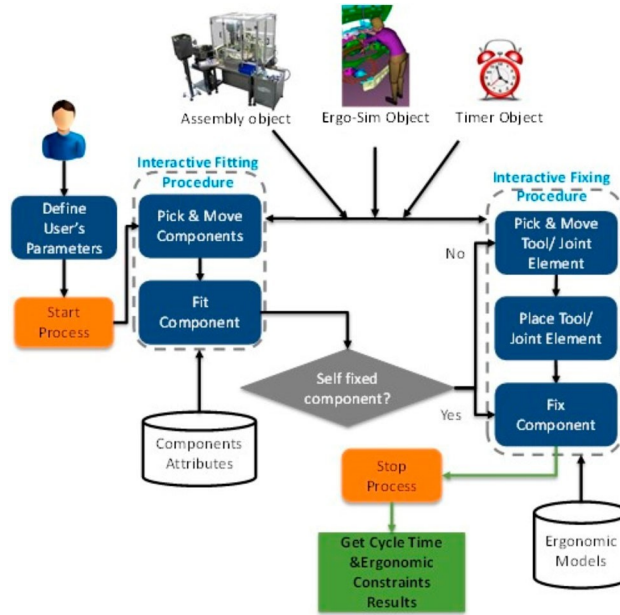


Figure 8. The concept of the user workflow in the virtual environment (Chryssolouris et al. 2004).

3.2.8. Ergonomics simulation

Ergonomics are defined as ‘the theoretical and fundamental understanding of human behavior and performance in purposeful interacting socio-technical systems, and the application of that understanding in order to design of interactions in the context of real settings’ (Wilson 2000). The human involvement is very critical, as it influences the feasibility, the cycle time, the working comfort and the safety of an operation (Chryssolouris et al. 2004). Emerging technologies support simulation-based engineering to address this in a proactive manner allowing workplaces and tasks to be simulated (Brolin et al. 2018). The comparison between ergonomic measurements in virtual and real environments during some specific task is presented in (Hu et al. 2011). Kara et al. (2014) suggest an integrated model that incorporates ergonomics and resource restrictions into assembly line balancing. The ErgoToolkit platform implements ergonomic analysis methods, already available in the literature or company practice, in digital tools for ergonomics, namely, Posture Definition and Recognition, and Stress Screening, integrated into state-of-the-art virtual manufacturing software (Alexopoulos, Mavrikios, and Chryssolouris 2013). An approach to human motion analysis and modelling with respect to anthropometric parameters is tested by Mavrikios et al. (2007).

As systems’ complexity and the number of features are only increasing, novel ways of effectively implementing them should be explored. The interaction between models is required, as well, for an integrated approach for addressing daily design problems, considering the integration of models into an encompassing DHM (Digital Human Modelling) (Jayaram et al. 2006). Moreover, data harmonisation is crucial so as to develop integrated models (Moes 2010) (Figure 8).

3.2.9. Manufacturing execution systems

A Manufacturing Execution System (MES) is a system that supports manufacturers to attain constant product quality, comply with regulatory requirements, reduce time to market, and lower production costs (Larreina et al. 2013), with an emphasis on total quality management programmes.

An IDEF0-model of an ‘order handling’ shop floor based on FMS lines is developed by (Choi and Kim 2002) to identify functional requirements of MESs, and then a two-tier MES architecture satisfying the functional requirements. A holonic MES that utilises a given schedule as a guideline for selecting among task execution alternatives, which are independent of the original schedule, is proposed in (Valckenaers et al. 2007). An adaptive decision-making system that supports distributed execution control and emergent scheduling is presented in (Rolón and Martínez 2012). A Radio Frequency Identification (RFID)-enabled real-time manufacturing execution system is proposed in (Zhong et al. 2013); RFID devices are used in order to track and trace manufacturing objects and acquire real-time production data, identification, and control of disturbances. Manufacturing Execution Systems in the future need to support extended horizon of planning (Valckenaers et al. 2007).

3.2.10. Supervisory control and data acquisition

Supervisory Control and Data Acquisition (SCADA) is the technology that enables data collection from one or more distant facilities while sending limited control instructions to those facilities (Boyer 2004). A methodology for deriving main scheduling decisions is introduced by Mourtzis et al. (2014) for exploitation of machine information in near real-time machine condition monitoring.

A significant step of SCADA and System Under Control is a hybrid modelling and simulation approach (Jahangirian et al. 2010), which is used to realise the identification and assessment of hidden vulnerabilities (elaborated more in the corresponding section). Son (2011) presents a solution and a design procedure of supervisory control problem for the case of decentralised control and applies the proposed design procedure to an experimental miniature computer-integrated manufacturing system. A state of the market review is presented in (ReportsnReports 2014). Whilst SCADA systems have already been integrated at a large portion in manufacturing, it is important to be supported with increased rates of data exchange with external ICT tools and as a result, with improved practices for data security and integrity (Nicholson et al. 2012). Towards this direction, Mousavi and Siervo (2017) presented a platform for real-time data acquisition combined with discrete event simulation to accurately measure key manufacturing systems performance metrics for enabling the performance optimisation and destabilising factors' prediction.

3.2.11. Supply chain simulation

A supply chain is a value-adding chain of processes from the initial raw materials to the ultimate consumption of the finished product, spanning across multiple supplier-customer links (Dugal, Healy, and Tankenton 1994). Modern manufacturing enterprises must collaborate with their business partners through their business process operations such as design, manufacture, distribution, and after-sales service (Mourtzis 2011). The principles of the supply chain control logic are presented in (Makris, Zoupas, and Chrysosolouris 2011).

A generic hybrid-modelling framework for supply chain simulation is presented in (Umeda and Zhang 2010) and the corresponding study performed in a Tesla case-study (Gianesello, Ivanov, and Battini 2017). A method to model, simulate and optimise supply chain operations by taking into consideration their end-of-life operations is used by Komoto et al. (2011) to evaluate the capability of OEMs to achieve quantitative performance targets. Moreover, a model was implemented in the form of an Internet-enabled software framework, offering a set of characteristics including virtual organisation, scheduling, and monitoring, in order to support cooperation and flexible planning and monitoring across extended manufacturing enterprise (Mourtzis 2011). In addition to that, the application of the simulation approach to a supplier selection example is illustrated by Mourtzis et al. (2016).

In the future, identifying the benefits of collaboration and partnership and overcoming the barriers in supply chain simulation will support healthy collaborative arrangements (Ramanathan 2014).

3.2.12. Design and planning of manufacturing networks

A modern manufacturing network is composed of cooperating OEM plants, suppliers, and dealers that produce and deliver final products to the market (Mourtzis, Doukas, and Psarommatis 2013b). The evaluation of the performance of automotive manufacturing networks under highly diversified product demand has been examined through discrete-event simulation models with the use of multiple conflicting user-defined criteria such as lead time, final product cost, flexibility, annual production volume, and environmental impact due to product transportation (Mourtzis, Doukas, and Psarommatis 2013b). A method implemented on a software tool that comprises a mechanism for the generation and evaluation of manufacturing network alternative configurations is demonstrated in (Mourtzis, Doukas, and Psarommatis 2013a). The complexity and the stability of MS are investigated in (Papakostas et al. 2009; Efthymiou et al. 2014) by introducing concepts based on discrete event simulation and nonlinear dynamics theory, and in (Mourtzis, Doukas, and Psarommatis 2013a), using the Lempel-Ziv algorithm for dynamic complexity assessment. A stochastic network design problem addressed in (Salem and Haouari 2017), though a simulation-optimisation approach, which aims to modify the expected quantities that are ordered to cope with the randomness of supply and demand parameters in a globalised economy. An interesting review of available solutions is presented here (Gupta, Singh, and Verma 2010) (Figure 9).

As production networks become globalised and increasingly complex, the need for solutions that may tackle the design of MS increases. Discrete event simulation software, metaheuristics, queuing theory, methods from operations research, and mathematical programming, are commonly used to global manufacturing networks (Hochdörffer et al. 2018). Existing platforms often fail to tackle the numerous issues of manufacturing network management in a holistic integrated manner, because they refer to not directly related manufacturing information and context (e.g. long-term strategic scheduling vs. short-term operational scheduling) (Si et al. 2011). The harmonisation, both on an input/ output level and to the actual

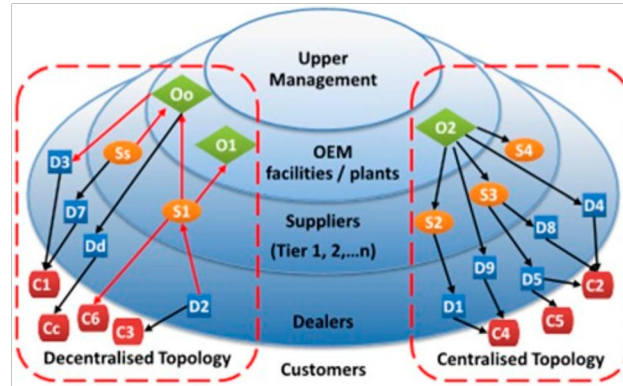


Figure 9. Decentralised manufacturing networks topologies (Mourtzis, Doukas, and Bernidaki 2014).

contents of information is a necessity in future development, which hinders the applicability of tools to real life MS and reduces the trustworthiness of results in the eyes of the end-users.

3.3. Digitalisation in simulation- towards the 4th industrial revolution

As digitalisation is reshaping manufacturing, from product design and production to lifecycle assessment, emerging technologies have increased impact in simulation (Mourtzis, Papakostas, et al. 2012). The industry has started to integrate advanced approaches towards interconnected systems that support higher precision decision making and realistic representation of the entities in the digital world (Chrysosouris, Mavrikios, et al. 2008).

The 4th industrial revolution introduces technologies such as virtual and augmented reality, hybrid simulation and digital twin. The first two support advanced realistic visualisation, simulating design and functionality of existing entities in an intriguing way that promotes knowledge sharing and intuition. Virtual prototyping provides a new step in product design, prior to materialisation (Azuma et al. 2001). As the production models become more and more complex, hybrid simulation becomes a highly used approach (Saouma and Sivaselvan 2014). The last few years, digital twin has arisen as a technology that combines data-driven real-time monitoring of real entities with scenarios simulation, aiming to improve its functionality and maximise its efficiency (Tuegel et al. 2011). Digital twin is not solely focused on one entity but may be expanded to whole systems, thus creating a holistic approach for the digitalised factories of the future.

3.3.1. Augmented reality

Augmented Reality (AR) is defined as 'a real-time direct or indirect view of a physical real-world environment that has been enhanced/ augmented by adding virtual computer-generated information to it' (Azuma et al. 2001). The challenge is to design and implement integrated AR-assisted MS that could enhance the manufacturing processes, as well as product and process development (Nee et al. 2012). In addition, the exploitation of AR environment in the rapid creation and modification of freeform surfaces is introduced, and methods for enabling increased flexibility during exploratory, conceptual industrial product design through three-dimensional (3D) sketch-based user input are explored (Fuge et al. 2012). Fang, Ong, and Nee (2012) aimed to facilitate robot programming and trajectory planning with the use of an AR-based system. In order to enhance the perception of the customer towards a personalised product design, product customisation, and personalisation applications have emerged (Mourtzis, Doukas, and Psarommatis 2012; Mourtzis et al. 2017).

Huang, Ong, and Nee (2015) presented a system supported by a wireless sensor network, which integrates sensor measurement and real-time Finite Element Analysis simulation into an AR-based environment. In (Steyn et al. 2015) the design and modelling of an AR platform are discussed for disabled wheeled mobility studies. Indicatively, AR was applied through integrating a hybrid rendering of volume data, vision-based calibration, accurate real-time tracking methods, tangible interfaces, multimedia annotations, and distributed computation and communications at Daimler-Chrysler (Baratoff and Regenbrecht 2004). Moreover, in (Park et al. 2015) a new AR application is presented, which is based on projection, namely Spatial Augmented Reality. Applications that connect production machinery monitoring and remote maintenance support are reported in (Mourtzis et al. 2017). From a market perspective, a review of available solutions is presented in (MarketsandMarkets 2017). To secure future longevity in manufacturing, AR applications need increased accuracy and mobility while reducing the mental workload for the user; a challenge both from hardware (lighter mobile computational devices, improved battery life, and optics) and software (more accurate positioning algorithms) (Nee et al.

2012). Additionally, connecting the digital information from other systems (e.g. PLM tools) directly to AR applications, thus reducing the required expertise and time, will expand the use of Augmented Reality in the 4th Industrial revolution era.

Though relatively young, the market of AR software is highly growing with toolkits that target different devices (AR headsets, mobile devices). ARToolKit set the ground as a software for the first implementations in the research environment using AR (Kato, Billinghurst, and Poupyrev 2000). Next, commercial platforms that accelerated the implantation, such as Vuforia,¹ brought the applications in industrial and commercial applications. As the competition rises and new features are implemented (object recognition, specific sensor data integration) new software platforms that target specific devices or operating systems arise, such as ARKit and ARCore for mobile devices, or Meta for specific headsets (Xu et al. 2018). Platforms that target manufacturing applications, such as Reflekt One, are also making their first steps, but are not yet widely used.²

3.3.2. Virtual reality

Virtual Reality (VR) is defined as the use of computers and other special hardware and software to generate the simulation of an alternative world or environment in real-time, believable as real or true by the users. VR is a rapidly developing computer interface that strives to immerse the user completely within an experimental simulation, thereby enhancing the overall impact, and providing a much more intuitive link between the computer and the human participants (Lu, Shpitalni, and Gadh 1999). VR technology allows engineers/designers to interact with the 3D model in an immersive environment and enables testing, experimentation, and evaluation of the product in full context (Rentzos et al. 2014). VR has found application from the design to the process simulation phase (Mujber, Szecsi, and Hashmi 2004). Recently, new semantic-based techniques are introduced in order to facilitate the design and review of prototypes by providing usability and flexibility to the engineer/ designer (Makris, Rentzos, et al. 2012).

In the field of collaborative management and verification of design knowledge, VR allows multiple users to be immersed and interact with the virtual prototype, efficiently managing the knowledge during the product design phase, collaborating in real-time on the same virtual object, while making ergonomic evaluation with the use of digital human simulation (Chrysosolouris, Papakostas, and Mavrikios 2008). Additionally, VR is also exploited for workplace design and analysis (Michalos et al. 2018). Furthermore, Peng et al. (2007) propose a networked Virtual manufacturing system, which was composed of 3D dimensional CAD/CAM modelling, VR, production planning and control and simulation, to address knowledge management issues. Another haptic VR platform, HAMMS, was introduced in (Gonzalez-Badillo, Medellin-Castillo, and Lim 2013) in order to facilitate the performance, the planning and the evaluation of virtual assembly of components. VR market leaders can be found in (MarketsandMarkets 2017). Last but not least, analysis of human-related aspects can be evaluated through VR to ensure workers' well-being in a manufacturing plant (Barrett et al. 2018).

3.3.3. Hybrid simulation

A key approach to simulate complex systems that incubate both discrete event and continuous behaviour is the utilisation of hybrid simulation. In order not to compromise the quality of the final simulation model, a different approach in each sub-system and their propagation is used to simulate the whole system (Saouma and Sivaselvan 2014). Apart from their application in dynamic models, such as trajectory calculation and bipedal robots (Chevallereau, Grizzle, and Shih 2009), they are also applied at large in production simulation, solving complex systems that may shift between discrete and continuous models. Combining different methods to achieve hybrid approaches can be performed under different schemes; interesting taxonomies are presented in (Swinerd and McNaught 2012; Morgan, Howick, and Belton 2017). An approach that integrates hybrid to interconnect Things towards smart manufacturing is reported in (Nagadi et al. 2018).

Although hybrid simulation offers an intriguing approach to tackle complex problems, it still has some drawbacks. Although the concept was suggested early in the literature, the terminology and definitions are not yet consolidated (Balaban, Hester, and Diallo 2014) and the information sharing between the combined methods still follows per-case solutions (Gil-Garcia and Martinez-Moyano 2007). Additionally, the lack of formalisation in method selection is a great challenge for implementations to reach higher levels of maturity (Barbosa and Azevedo 2017). An overview of available software packages that support hybrid simulation approaches is presented in (Mosterman 1999).

3.3.4. Digital twin

Digital twin is a relatively young approach in manufacturing simulation (Tuegel et al. 2011). NASA also integrated it in a roadmap report of modelling and simulation technologies, defining it as a top challenge for the years to come, also providing a definition for it: 'A Digital Twin is an integrated multiphysics, multiscale, probabilistic simulation of an as-

built vehicle or system that uses the best available physical models, sensor updates, fleet history, etc., to mirror the life of its corresponding flying twin' (NASA 2012). In the digital twin approach, the physical entity, varying from a single resource to a whole product, is connected to a set of sensors and its status is monitored in real-time. Through simulation, the data are transformed into meaningful knowledge, interchangeable between the digital and the physical entity (Wasserman 2017). The digital twin of a product, begins from its design and manufacturing phase, storing important data from its creation, though most data are generated in its operation phase; an interesting review on the impact of each phase is presented by Boschert and Rosen (2016). As Cyber-Physical Production Systems become a trend in manufacturing (Mourtzis et al. 2016), digital twin integration for simulating their function and securing their longevity is inevitable (Rosen, Wilchert, and Bettenhausen 2015).

Although digital twin is a highly promising technology that can highly affect digitalised factories of the future, it is not yet mature nor is its full capacity explored in the literature, as it requires the integration of more than one technology in the areas of application, such as machine sensing, monitoring systems, and historical experience databases. The integration will enable the entities, on the one hand, to become connected to the Industrial Internet of Things while also providing them increased autonomy, in terms of human interference (programming, corrections) required. The integration of the technologies that will render the digital twin capable of accurate behaviour prediction, such as data integration and machine learning is crucial.

4. State of the market of simulation tools in the 4th industrial revolution

As technology rushes forward to the new era of integration of Internet of Things in manufacturing, so does the software solutions that aim to support the design and operation of such systems. The new era is characterised by an increased amount of data to be fail in real-time and feed simulation models that can quickly investigate different scenarios and provide decision support and optimisation of the system operation.



Figure 10. Major identified gaps for each product and production lifecycle sector.

Towards that end, large software companies are providing solutions that integrate the aforementioned Industry 4.0 technologies in unified platforms, benefiting also from machine learning techniques. Watson is a unified platform that implements digital twin applications with augmented reality simulation of operation.³ Azure IoT Hub⁴ is another platform that gains a lot of market share, due to its integration with other digital solutions that had already reached high levels of

industrial integration, while offering a connection with advanced machine learning interfaces. MindSphere⁵ is an IoT-based operating system that offers open integration interfaces for new developments and connection with digital twin simulation software. The aforementioned tools are not cut off their roots of basic simulations in manufacturing, such as a CAD, CAM or production simulation, but bundle them with IoT data connection, cloud services, and digital twin features, expanding their capabilities for the new era.

Other platforms have also emerged the last few years, by providers that did not provide simulation software in the past but become directly active in the new technologies (e.g. augmented reality), offering competitive solutions; ThingWorx.⁶ As more and more technologies will allow the modelling and simulation of MS, integrating data from various sources and feeding them to more complex and more realistic models (digital twins).

5. Conclusions- discussion

Digital manufacturing technologies are considered an imperative set of support tools in the continuous effort towards the reduction in the development time and cost of a product, as well as towards the expansion in product customisation options. The simulation-based technologies constitute a focal point of digital manufacturing solutions since they allow the experimentation and validation of different product, process, and manufacturing system configurations. The simulation tools reviewed in this research are constantly evolving and they certainly lead towards more efficient MS. But, in the current highly competitive business environment, which is constantly facing new challenges, there is always a need for even more efficient and adaptive technologies. Hereafter, the identification of the major gaps of each simulation approach in the aforementioned product and production lifecycle tools are discussed and future trends are outlined.

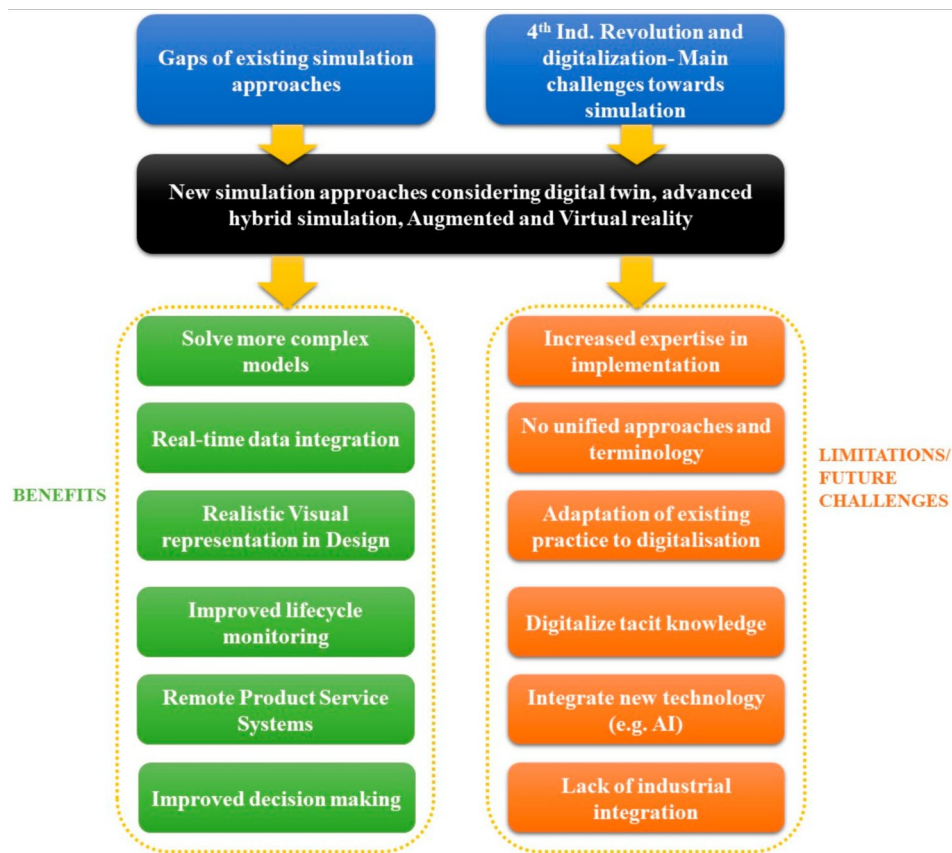


Figure 11. Benefits, limitations and future challenges posed by the new simulation approaches.

As a general conclusion deriving from the review and identified gaps (Figure 10), it can be said that, undeniably, the evolution of simulation tools and technologies has been huge during the last decades, yet undoubtedly the area is and will remain a fertile field of research and applications.

As digitalisation offers a solution to a great deal of the aforementioned limitations, it also raises new problems that need to be addressed. Figure 11 below summarises the benefits that the digitalisation technologies will bring to manufacturing, also highlighting the challenges that rise from their integration.

Concluding, digitalisation offers a set of new technologies and data that will assist the transition of product design and manufacturing simulation to the new era. Integrating these technologies in the current practice, while overcoming the obstacles, is a challenge for the academia and the industry. As data constitutes the fuel of industrial Internet of Things, simulation can be proven the enabler towards digital integration and access to digital data across the manufacturing life cycle in the future factory. The virtual testing of plant production systems at the planning phase, for any false prevention, arises also as a future application of simulation in MS (Deloitte 2018). Major concerns on worker safety and efficiency though, during their training period launch simulation of an entire factory or warehouse as an emerging trend. Additive manufacturing paired with advanced technologies from IoT can empower advanced analytics towards a more effective design cycle with a boost on the product's time-to-market. The factory of the future embraces the virtual factory concept which imposes change from predicting how the product is to be manufactured into how the product is actually being manufactured through real-time simulation. Potential that is generated from the abovementioned concept lies under factory assets' monitoring by analysing historical data in order to forecast failure, quality monitoring and all these functions in realtime. Briefly, the tendency and the opportunities are born through the simulation in the digitalised era of manufacturing, is directed to the creation of an operational view of the entire factory floor where access is possible for anyone, at any time, from anywhere (Aberdeen Group 2017). Under this concept, the design and the operation of the MS are driven by a plenty amount of meaningful data leading to better state inspection as so to better decisions in any phase of the product's lifecycle. Though, this is an opportunity for higher intuition in simulation, in the ever-going effort to approach the real model as accurately as possible.

Notes

1. <https://www.vuforia.com/>
2. <https://www.re-flekt.com/reflekt-one>
3. <https://www.ibm.com/internet-of-things/trending/digital-twin>
4. <https://azure.microsoft.com/en-us/services/iot-hub/>
5. <https://www.siemens.com/global/en/home/products/software/mindsphere.html>
6. <https://www.ptc.com/en/products/iot/thingworx-platform>

Disclosure statement

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